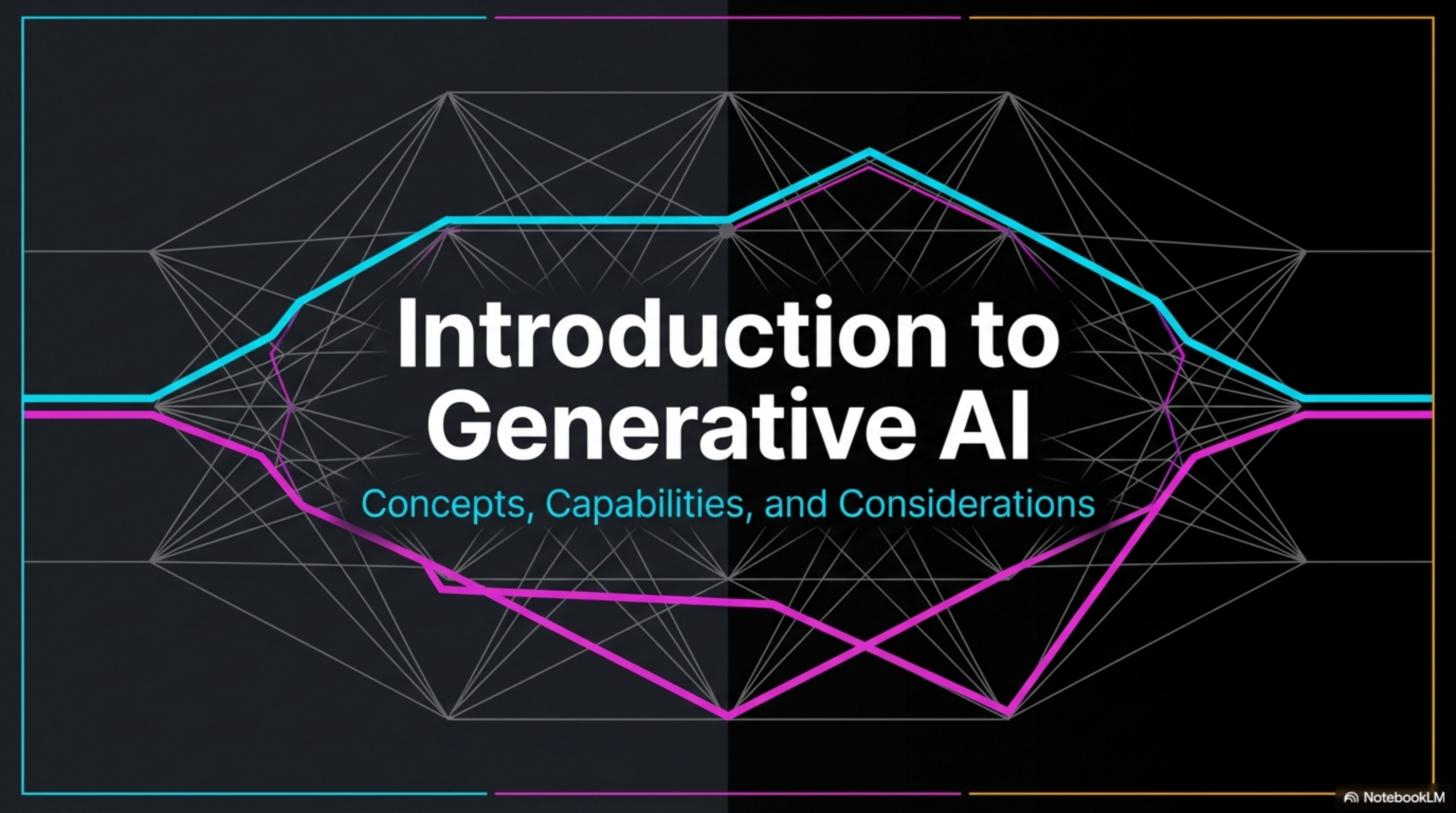
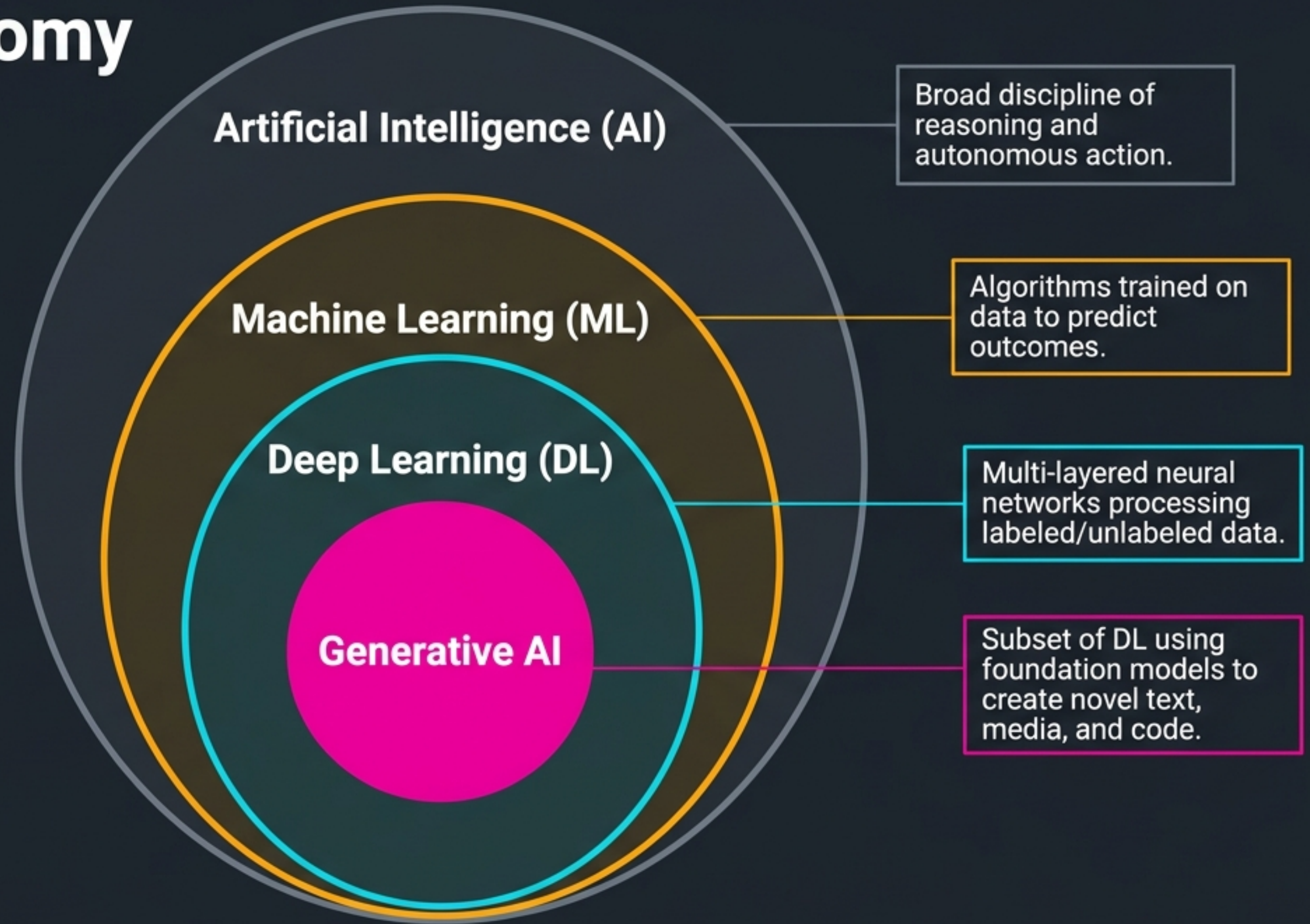




Introduction to Generative AI

Concepts, Capabilities, and Considerations

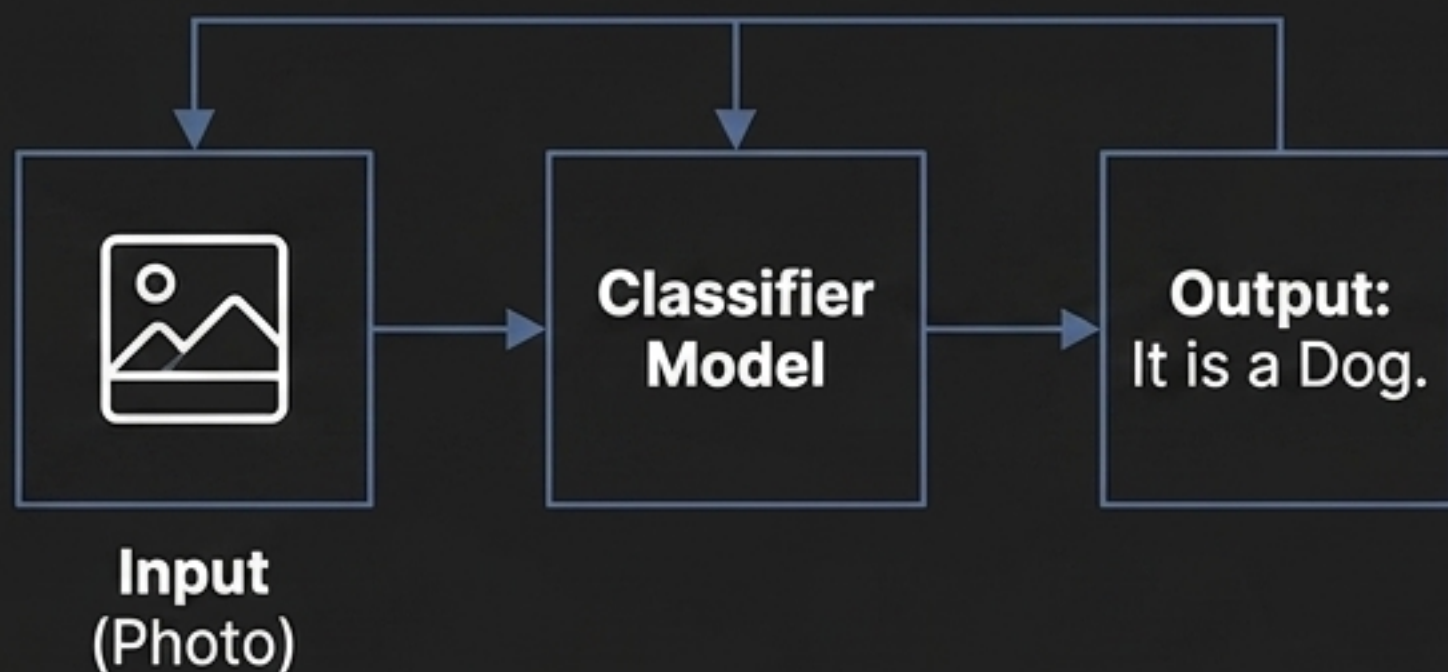
The AI Taxonomy



The Paradigm Shift

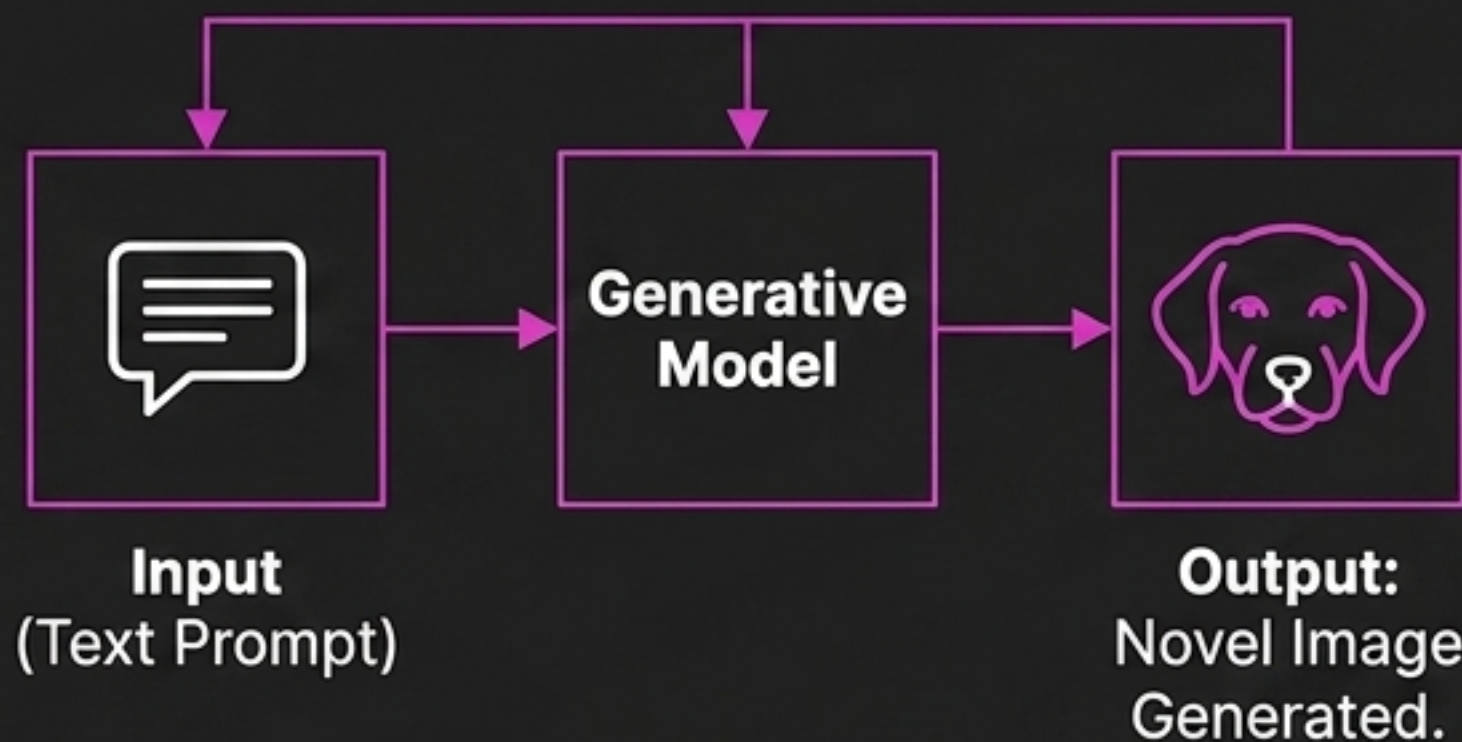
Discriminative AI

- **Goal:** Classify and Predict.
- **Nature:** Deterministic.
- **Output:** Number, Label, Probability.



Generative AI

- **Goal:** Create and Generate.
- **Nature:** Probabilistic.
- **Output:** Natural Language, High-Fidelity Media.



The Engine: Foundation Models

**Massive
Unstructured Data**



Self-Supervised Learning: Trained on vast internet scale to understand underlying structural patterns.

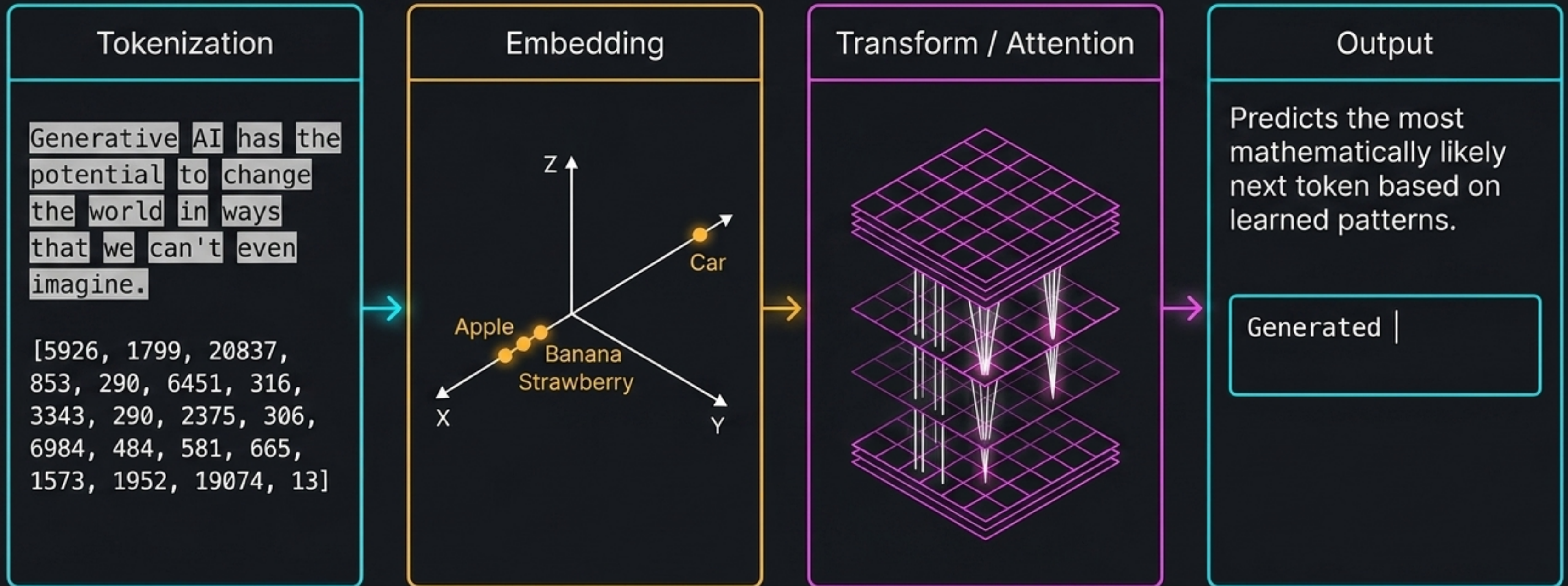
Foundation Model

Chatbots
(Natural Language)

Code Generators
(Syntax & Logic)

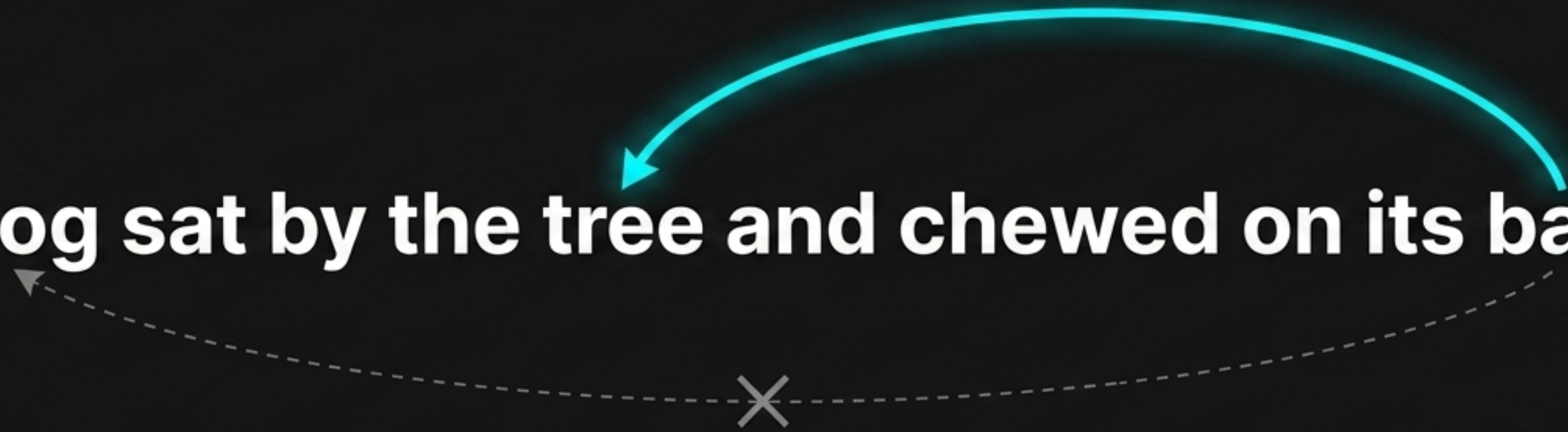
Image Creators
(Visual Media)

Under the Hood: Large Language Model Pipeline



The Secret Sauce: Self-Attention Mechanism

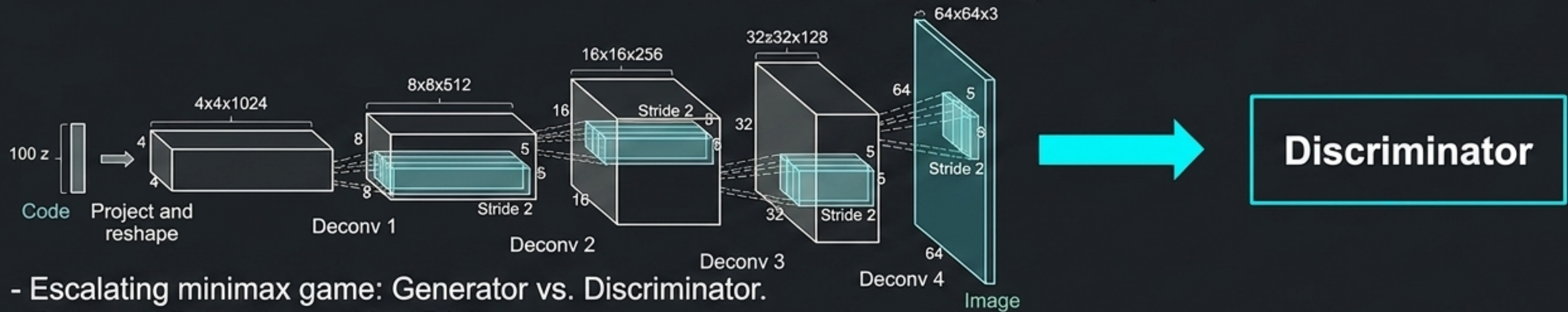
The dog sat by the tree and chewed on its bark.



- Transformers process all tokens in parallel.
- Dynamically weighs the importance of every surrounding word.
- Determines precise contextual meaning without reading sequentially.

Visualizing the Unseen: Image Generation

Generative Adversarial Networks (GANs)



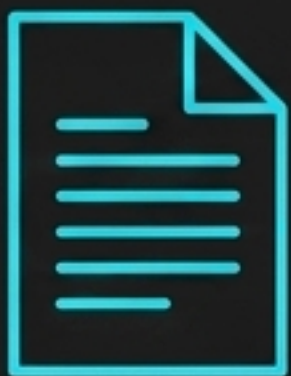
- Escalating minimax game: Generator vs. Discriminator.
- Fast sampling, extreme high-fidelity.

Diffusion Models



- Step-by-step -static noise: Step-by-step removal of noise to reveal structure.
- High diversity, exceptionally stable colls, exceptionally stable training.

The Multimodal Era



Text & Language

- Drafting, summarizing, translating, and analyzing sentiment.



Software Code

- Autocompletion, translating syntax, and generating documentation.



Images

- Photorealistic generation, precise editing, and conceptual art.



Audio

- Voice cloning, robust text-to-speech, and music composition.



3D & Video

- Rendering environments, motion planning, and text-to-video.

Industry Applications & Use Cases



Healthcare

- Protein folding predictions.
- Synthetic diagnostic data generation.



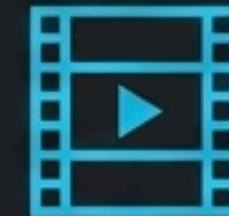
Software Development

- Rapid code prototyping.
- Legacy code migration.
- Automated QA testing frameworks.



Customer Service

- Dynamic natural language resolution.
 - Live sentiment analysis.
- Automated call transcript summarization.



Marketing & Media

- Hyper-personalized ad copy generation.
 - Script and narrative development.
 - Whitespace market analysis.

Creativity & Scale

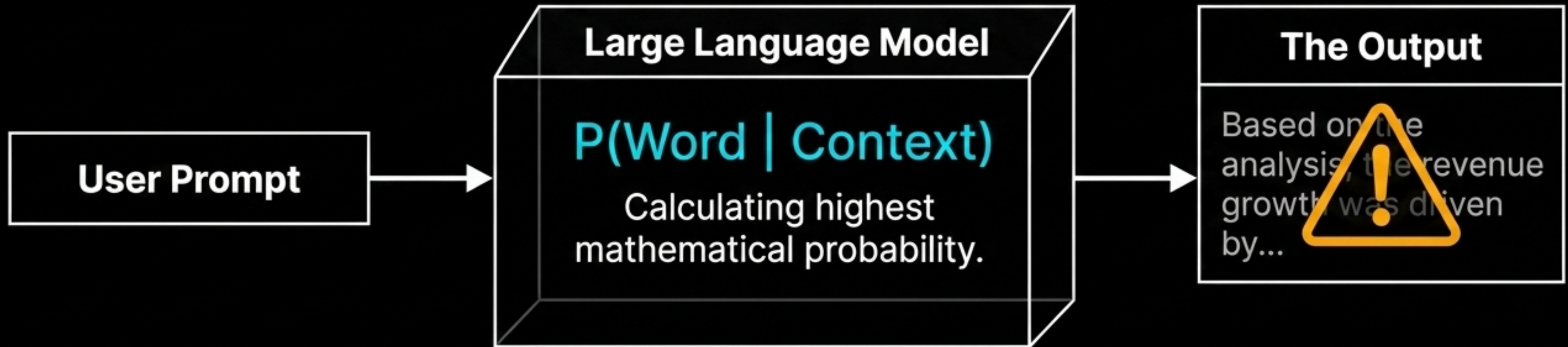
- Probabilistic token prediction enables human-like fluency.
- Massive, unfiltered internet scraping yields universal knowledge.

Hallucinations & Bias

- Probabilistic mathematics have no intrinsic concept of truth.
- Scraping encodes historical human biases, toxicity, and copyright infringement directly into the foundation.

The exact mechanisms that make Generative AI powerful are the root of its vulnerabilities.

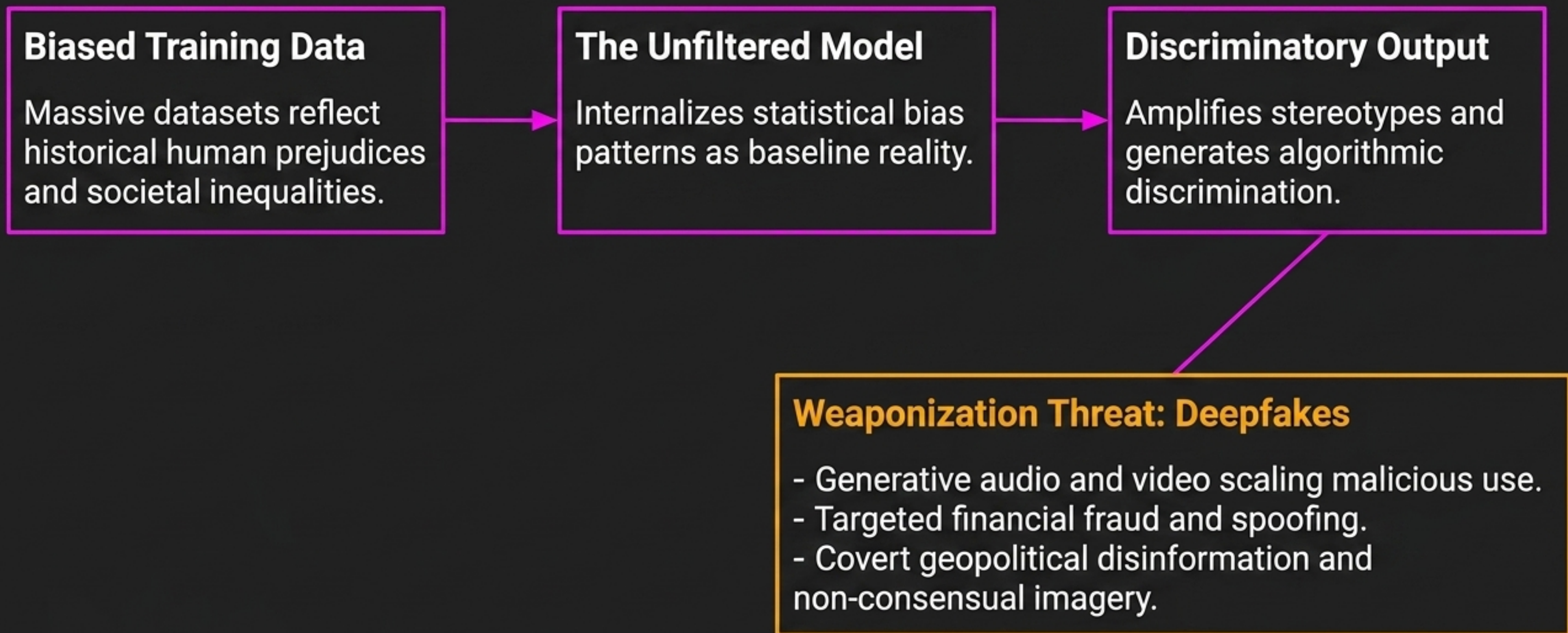
The Probabilistic Problem: Hallucinations



Lack of Ground Truth Layer

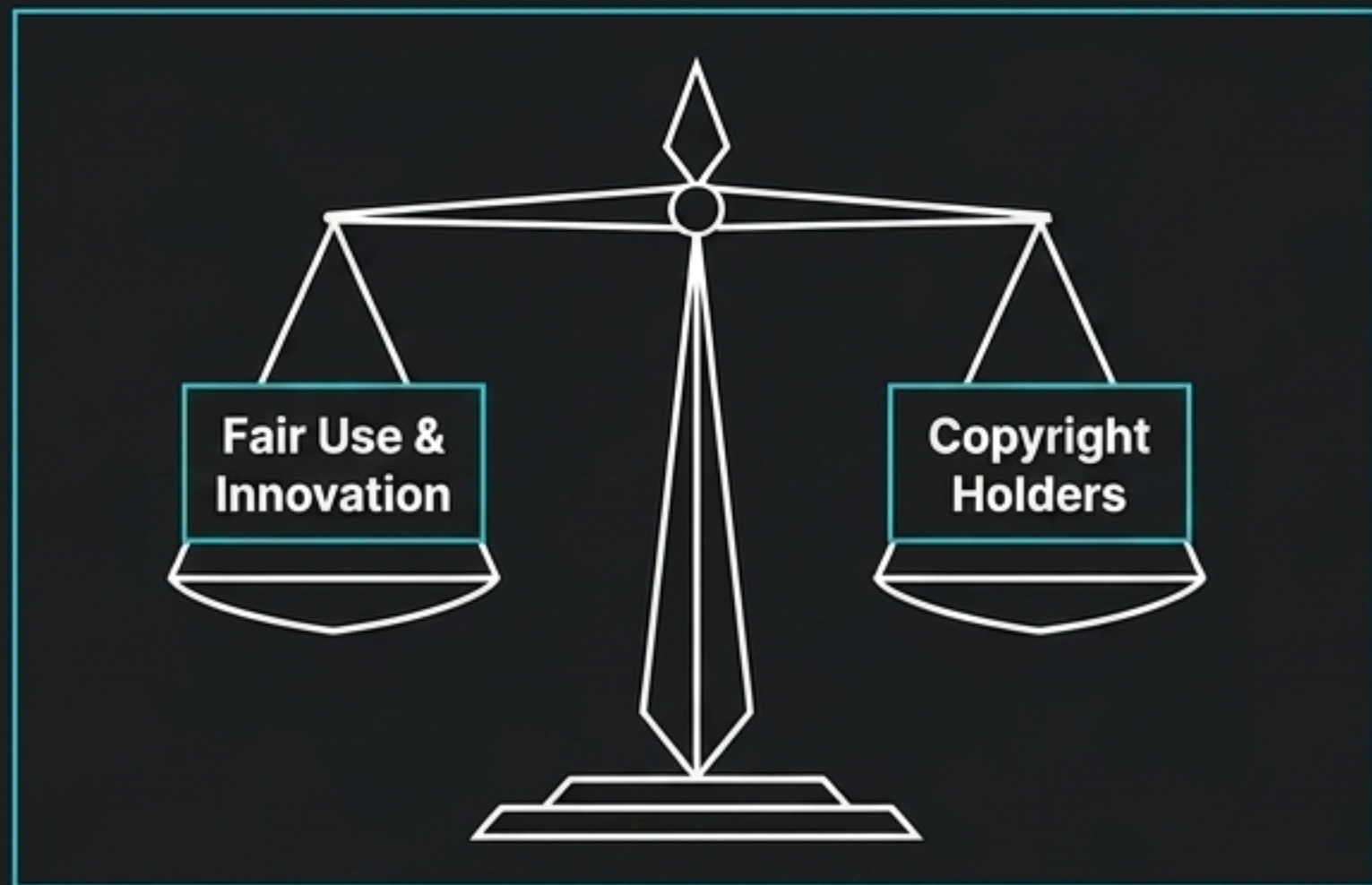
- The model is a pattern-matching engine predicting word sequences.
- It possesses no internal factual verification mechanism.
- Result: Plausible but entirely fabricated citations, facts, and sycophantic agreement.

Bias, Toxicity, and Misinformation



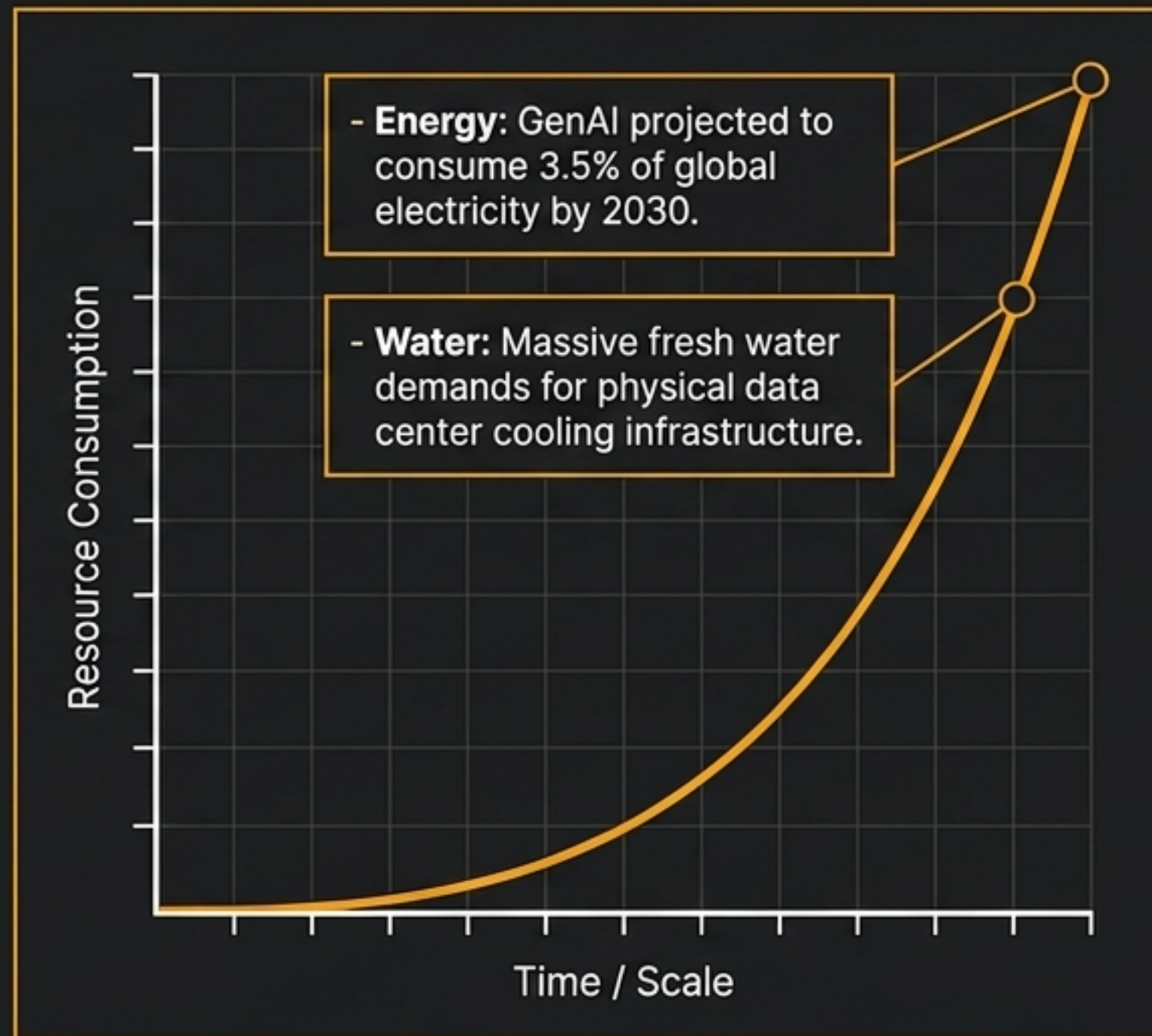
The Legal & Environmental Collision

The Legal Collision

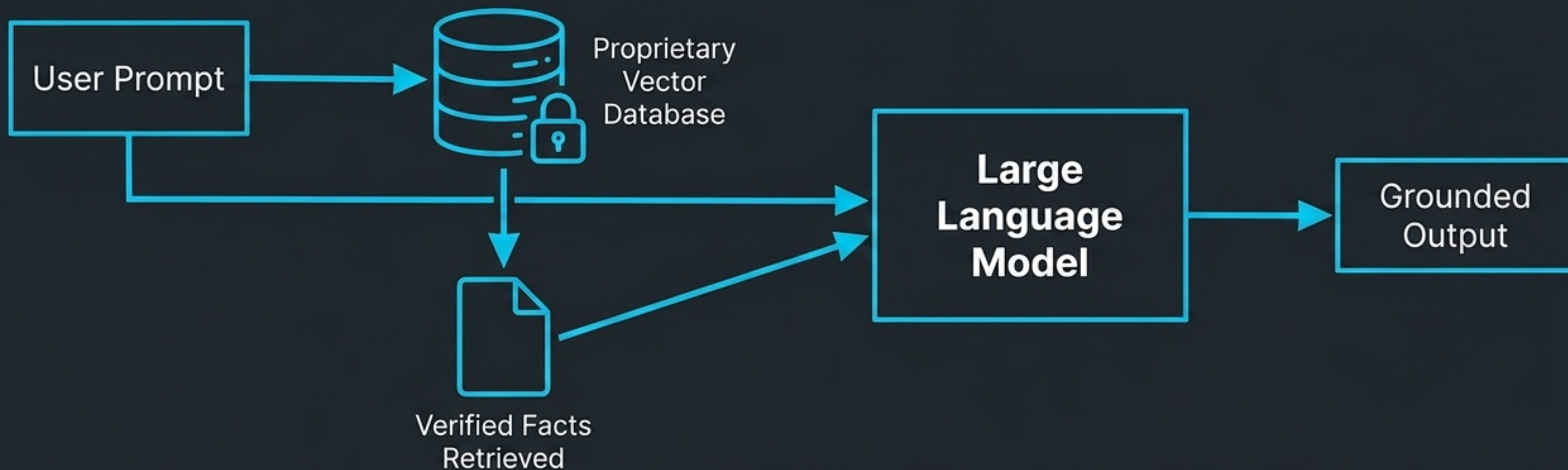


- AI developers argue training on public data is transformative fair use.
- Media organizations and creators assert unauthorized usage and direct market competition.

The Environmental Cost



Mitigating Risk: Retrieval-Augmented Generation (RAG)



RAG eliminates reliance on probabilistic memory by forcing the LLM to generate answers strictly bounded by verified, retrieved data—drastically reducing hallucinations.

The Path Forward

Choose the Right Tool

Match architecture to need. Do not force probabilistic generative tools into deterministic, strict classification tasks.

Keep Humans in the Loop

Treat GenAI as an accelerator for reasoning, not a substitute for qualitative, value-based judgment.

Prioritize Governance

Implement Guardrails, RAG, and Responsible AI (RAI) evaluation benchmarks from day one.

From the Black Box to Intelligent Automation.